

COURSE DESCRIPTION			
Program	Tool & Die Engineering		
Course Title	Basic CAD Lab		
Course Code	2119	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

This laboratory course provides hands-on training in computer aided drafting using CAD software. Students learn to use essential tools and commands for creating 2D drawings, orthographic and sectional views, and isometric drawings. The course also introduces basic concepts of 3D modelling and conversion of 3D models into 2D views to enhance drafting productivity and accuracy.

Course Objectives

1	To familiarize the different software used for drawing.
2	To introduce the basic commands and tools necessary for two-dimensional drawing, design and drafting using AutoCAD.
3	To develop skills for preparing orthographic, sectional and isometric drawings using CAD.
4	To introduce the basic concepts of 3D modelling in a software environment

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge about engineering drawing and Projections	1003	Engineering Graphics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Illustrate the use of computer aided drafting software.	Understand
CO2	Identify various commands used in CAD.	Understand
CO3	Apply knowledge to draw simple two-dimensional drawings and sections using CAD.	Apply
CO4	Construct isometric drawing of simple objects and identify basic concepts of 3D modelling.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	-	-	-	3	-	-	-	-	-	-
CO2	2	2	-	-	3	-	-	-	-	-	-
CO3	3	3	2	-	3	-	-	-	2	-	-
CO4	3	2	3	-	3	-	-	-	2	-	2
Total	10	7	5	-	12	-	-	-	4	-	2
Correlation Level	2.5	2.33	2.5	-	3	-	-	-	2	-	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Familiarize CAD software and its applications	Identify CAD software and its applications.	CO1	Understand	1
2	Identify applications, advantages and components of CAD	Explain applications, advantages and components of CAD.	CO1	Understand	1
3	AutoCAD interface and setting up a new drawing	Use AutoCAD window, start a new drawing and set units, limits, grid and snap.	CO1	Understand	1
Module II					
4	Coordinate systems and locating points	Use absolute, relative, polar and direct distance entry systems.	CO2	Understand	3
5	Draw and modify commands practice	Create 2D entities using draw and modify commands.	CO2	Understand	3
6	Dimensioning, text and plotting parameters	Apply dimensions, text and set plotting parameters.	CO2	Understand	3
	Series Exam - I		CO2	Understand	3
Module III					
7	2D entities: rectangle, polygon, circles, arcs, arrays and blocks	Create basic 2D drawings using layers, properties, hatch and blocks.	CO3	Apply	4
8	Orthographic views of simple components (minimum 5 figures)	Develop orthographic views with dimensions and text.	CO3	Apply	7
9	Sectional views of simple components (minimum 3 figures) and plotting	Create sectional views and plot/print drawings.	CO3	Apply	3
Module IV					
10	Isometric drawing using isometric snap and grid	Construct isometric views of simple objects with dimensioning.	CO4	Apply	4
11	Introduction to 3D modelling and solid primitives	Create simple 3D models using primitives and UCS/view controls.	CO4	Understand	3
12	Boolean operations and 3D commands (extrude, revolve, sweep)	Generate solids and modify using union/subtract/intersection.	CO4	Apply	2
13	Create 2D views from 3D object using BASE tool	Generate orthographic/isometric 2D views from 3D model in layout.	CO4	Apply	4
	Series Exam - II		CO4	Apply	3
	Open-ended experiment		CO4	Apply	3

Open-ended experiment**Objective:**

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
 Integrate multiple concepts from the syllabus into a combined application.
 Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
 Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Explore different CAD software used in engineering drafting and list their features	1.5

2	Watch a tutorial on AutoCAD interface and summarize key toolbars and command window.	1.5
3	Practice setting up a new drawing (units, limits, grid, snap).	1.5
Module II		
4	Practice coordinate entry methods with at least five simple exercises.	1.5
5	Prepare a short note on dimension styles and plotting settings	1.5
6	Practice draw and modify commands with simple objects.	1.5
7	Practice plotting/printing with correct scale and sheet size	1.5
Module III		
8	Create a mini drawing combining layers, hatch and blocks	1.5
9	Practice drawing one orthographic view set and one sectional view set.	1.5
10	Prepare a combined drawing sheet with orthographic + sectional views.	1.5
11	Develop one complete CAD drawing with layers, hatch, dimensions and text.	1.5
Module IV		
12	Practice isometric drawings of cube/step block/cylinder with dimensions.	1.5
13	Create a simple 3D model and generate 2D views using the BASE tool.	1.5
14	Practice Boolean operations (union, subtract, intersection).	1.5
15	Generate orthographic and isometric 2D views from a 3D object.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Engineering Graphics	P I Varghese, K C John	VIP Publishers

REFERENCE			
Sl. No.	Title	Author	Publication
1	Engineering Drawing	N. D Bhatt	Charotar Publishing House
2	Engineering Graphics with AutoCAD	Kulkarni, D. M.; Rastogi, A. P.; Sarkar, A. K.	PHI Learning
3	Essentials of Engineering Drawing and Graphics using AutoCAD	T. Jeyapoovan	PHI Learning
4	AutoCAD User Guide	Autodesk	Autodesk Press
5	AutoCAD for Engineers and Designers	Sham Tickoo	Dreamtech Press / Galgotia Publication

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	www.swayam.gov.in	All
2	https://nptel.ac.in/course.html	All
3	https://knowledge.autodesk.com/	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Desktop Computer Systems	Minimum i5 Processor / 8 GB RAM / 500 GB HDD or SSD	30
2	LAN and Networking Facility	For file sharing and lab connectivity	1 set
3	Projector / Smart Display	For demonstration of CAD commands and exercises	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	AutoCAD Software	Latest Version / Educational License	30

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10

Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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